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Exit Problems for Additive-Increase Multiplicative-Decrease Markov-Modulated Processes

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Joint work with **Zbigniew Palmowski** (Wrocław University of Science and Technology)

AIMD processes

AIMD means *Additive-Increase Multiplicative-Decrease*:
a positive process that grows gradually and is reduced proportionally at random decrease epochs.

additive increase	→	**random event**	→	**multiplicative decrease**
X grows linearly		loss, congestion, reset		$X \mapsto pX$

Pathwise intuition: upward crossings are continuous; downward crossings occur by jumps.

AIMD dynamics

Continuous-time model

$$\frac{dX_t}{dt} = \frac{1}{c}, \qquad T_n \le t < T_{n+1},$$

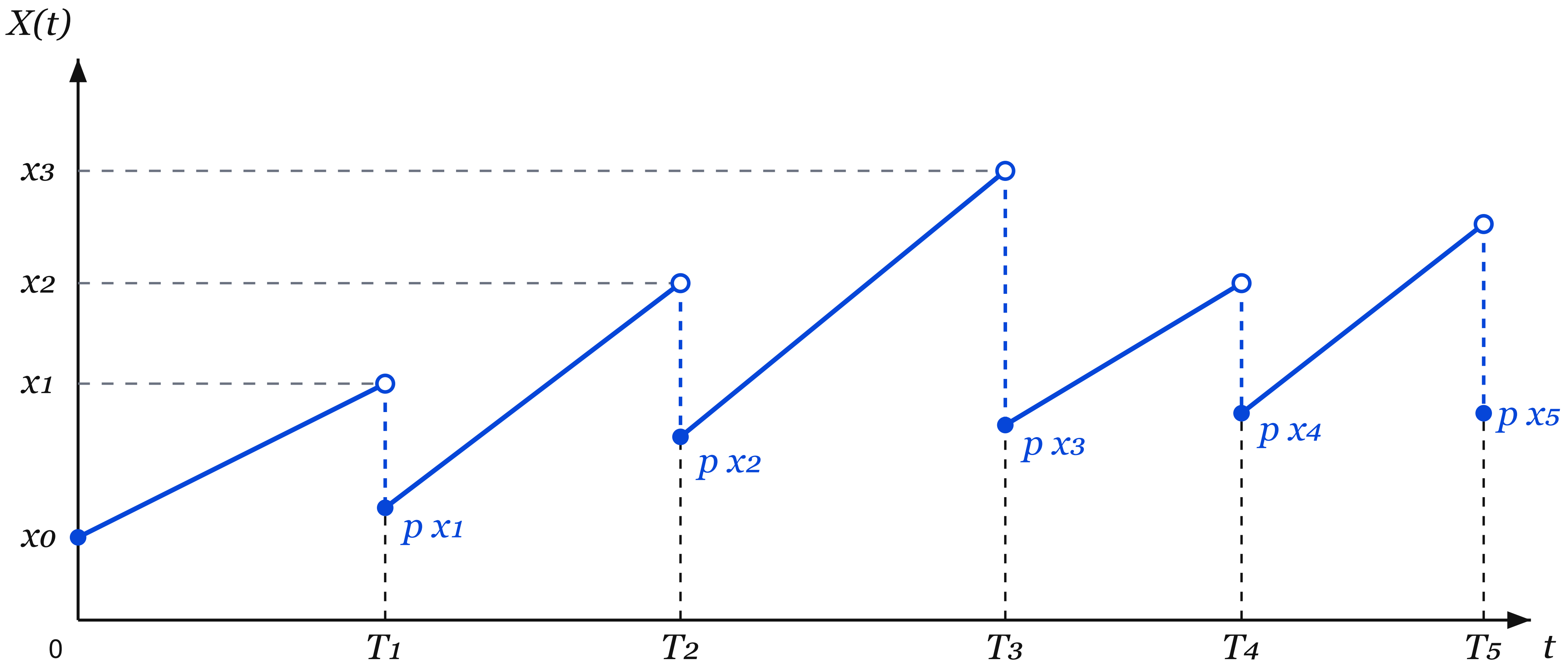
$$X(T_n) = pX(T_n-), \qquad 0 < p < 1.$$

A **Poisson clock** with rate λ drives the decrease epochs.

Discrete-time model

$$X_{n+1} = \begin{cases} X_n + 1/c, & \text{with prob. } (1-\delta), \\ pX_n, & \text{with prob. } \delta \end{cases} \qquad 0 < p < 1.$$

AIMD sample path



Additive Increase

Between events, the process grows linearly with constant rate $a > 0$.

Multiplicative Decrease

At random epochs T_k, the process jumps to a fraction of its current value, that is, $X(T_k) = p_k X(T_{k-})$, $0 < p_k < 1$.

AIMD and TCP

TCP congestion control protocol is among classical applications:
it cautiously increases the *transmission rate* when transmission succeeds,
to decrease it sharply when congestion is detected.

Successful transmission

$$W \longmapsto W + 1$$

Probe the available bandwidth by small additive steps.

Packet loss / congestion

$$W \longmapsto pW, \qquad p \approx \frac{1}{2}.$$

React quickly by reducing the current congestion window.

The same rule balances three goals: stability, fair bandwidth sharing, and adaptation to changing network conditions.

Earlier TCP/AIMD work

The TCP motivation led to a probabilistic literature on AIMD and growth-collapse dynamics.

Guillemin, Robert, Zwart (2004) *AIMD algorithms and exponential functionals*
Ann. Appl. Probab. 14(1), 90-117.

Boxma, Perry, Stadje, Zacks (2006) *A Markovian growth-collapse model*
Adv. in Appl. Probab. 38(1), 221-243.

van Leeuwaarden, Löpker, Ott (2009) *TCP and iso-stationary transformations*
Queueing Syst. 63, 459-475.

AIMD and Partial resetting

AIMD is one instance of a broader class of **processes with partial resetting**:
deterministic or stochastic growth interrupted by resets that keep part of the current state.

Search, restart and algorithms

Resetting appears in intermittent search and restart mechanisms, from abstract Markov processes to stress-release models.

Evans et al. (2020), Gupta and Jayannavar (2022), Bénichou et al. (2011), Avrachenkov et al. (2013), Borovkov and Vere-Jones (2000).

Queueing, clearing and growth-collapse

Queueing applications include catastrophe rates, random clearing events and growth-collapse models.

Artalejo et al. (2006), Kella et al. (2003), Boxma et al. (2006), Löpker and Stadje (2011).

more on Partial resetting

Biology, chemistry and ecology

Resetting ideas appear in RNA polymerase recovery, enzymatic reactions, molecular inhibition, pollination and ecological disturbances.

Grünwald and Singer (2010), Roldan et al. (2016), Berezhkovskii et al. (2016), Robin et al. (2018, 2019), Reuveni et al. (2014).

Physics, environment and risk

Related models are used in stochastic thermodynamics, quantum resetting, environmental jump processes, insurance and financial crashes.

Fuchs et al. (2016), Mukherjee et al. (2018), Dahlenburg et al. (2021), Mau et al. (2014), Suweis et al. (2010), Akahori et al. (2022), Sornette (2017).

Our starting point:

2023 AIMD scalar model

The study van der Hofstad et al. (2023)[†] looked at exit-time identities for the scalar model.

The model is specified by one triplet (c, λ, p):

- deterministic growth at rate $1/c$;
- Poisson jumps with rate λ;
- mult. decrease factor $p \in (0, 1)$.

that determines the dynamics of the X process:

$$\begin{aligned} dX_t &= dt/c, \qquad T_n \le t < T_{n+1}, \\ X(T_n) &= pX(T_n-), \\ T_{n+1} - T_n &\sim \mathrm{Exp}(\lambda). \end{aligned}$$

This talk - **Markovian modulation**

A finite-state Markov environment J modulates all local parameters.

$$(c_i, \lambda_i, p_i), \qquad \boldsymbol{Q} = (q_{ij}), \qquad i, j \in \{1, \dots, m\}.$$

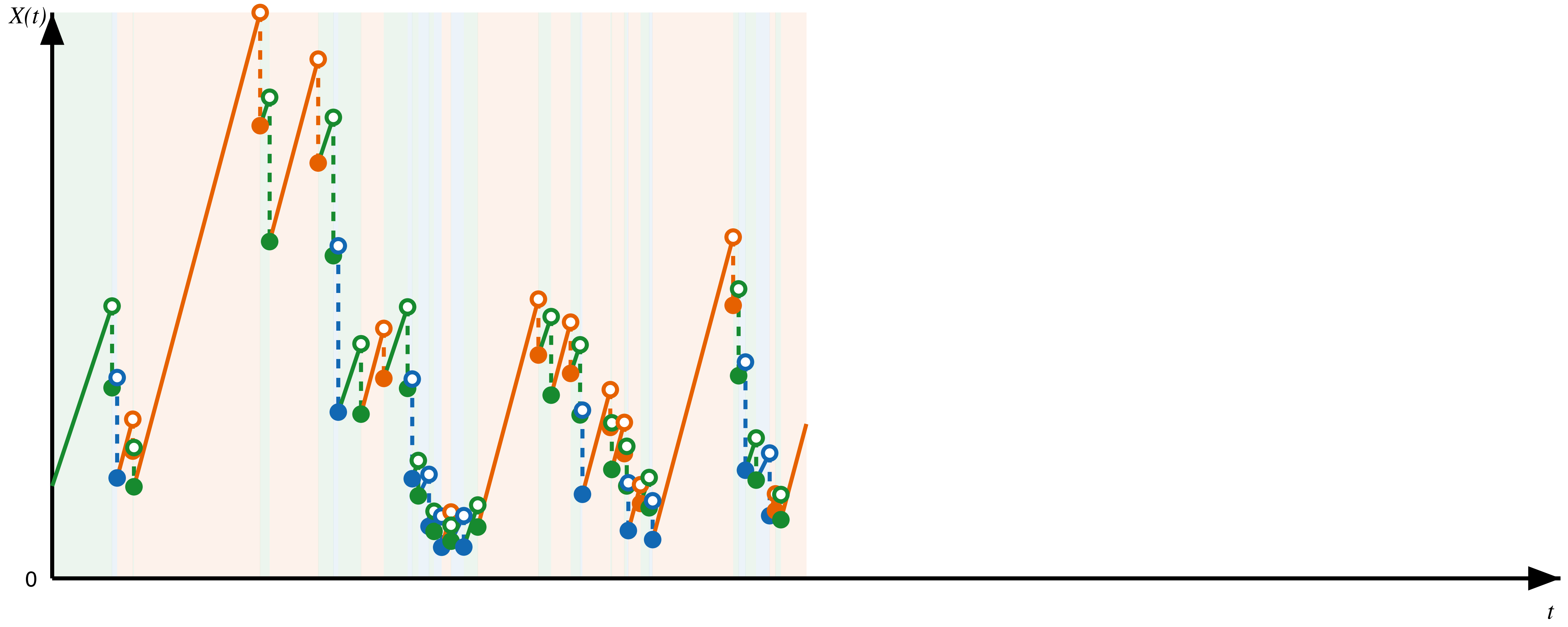
The scalar transforms of the exit-times become **matrix-valued** transforms, indexed by the starting and ending environment regime.

$$\begin{aligned} \{J(T_n) = i\} \implies \{J(T_{n+1}) = j\} &\text{ with prob. } q_{ij}, \\ dX_t = dt/c_i,\ J(t) = i &\qquad T_n \le t < T_{n+1} \\ X(T_n) &= p_i X(T_n-), \\ T_{n+1} - T_n &\sim \mathrm{Exp}(\lambda_i). \end{aligned}$$

Note: the skip-free/first-step strategy survives, but scalar q-series are replaced by non-commutative regime-indexed matrix series.

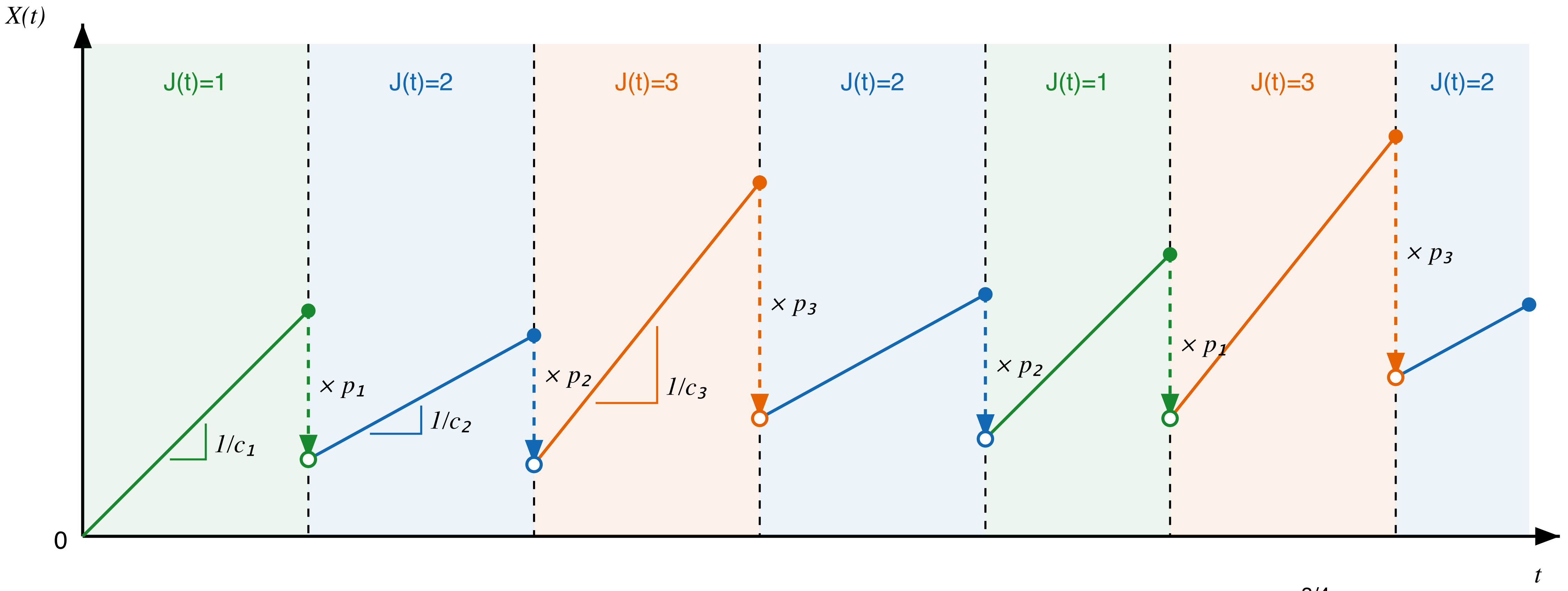
[†]R. van der Hofstad, S. Kapodistria, Z. Palmowski, S. Shneer. J. Appl. Probab. 60 (2023)

Markov modulated AIMD simulating one sample path

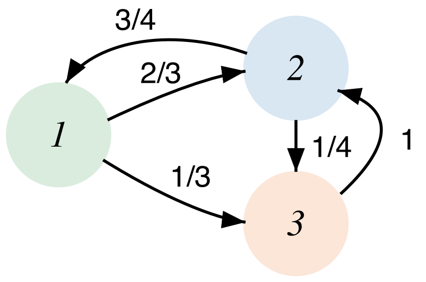


Hide future times

Markov modulated AIMD sample path



Whenever J(t)=i, the process increases linearly with fixed slope $1/c_i$.
At the jump time, the value of X(t-) is multiplied by p_i and the environment changes to one of the other states.
Dashed lines mark jump times; colors identify the active Markov environment.



Objects of interest

For fixed levels $0 \le b < a$, define

$$\tau_a^{\uparrow} = \inf\{t \ge 0 : X_t \ge a\}, \qquad \tau_b^{\downarrow} = \inf\{t \ge 0 : X_t \le b\}.$$

Upward one-sided transform $\boldsymbol{Z}(x \uparrow a)$

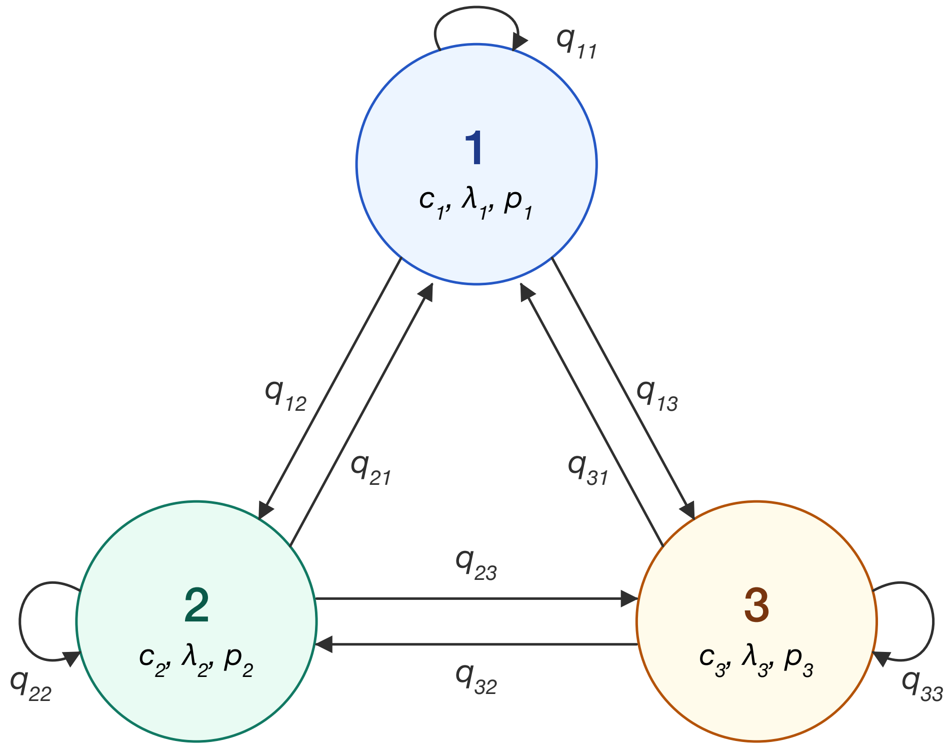
$$Z_{ij}(w, x \uparrow a) = \mathbb{E}_{x,i}\big[\mathrm{e}^{-w\tau_a^{\uparrow}};\ J(\tau_a^{\uparrow}) = j\big].$$

Two-sided upward transform $\boldsymbol{L}(x \uparrow_b^a)$

$$L_{ij}(w, x \uparrow_b^a) = \mathbb{E}_{x,i}\big[\mathrm{e}^{-w\tau_a^{\uparrow}};\ \tau_a^{\uparrow} < \tau_b^{\downarrow},\ J(\tau_a^{\uparrow}) = j\big].$$

Throughout, matrix products encode conditioning on the regime reached at an intermediate level.

A closer look at the introduction of the Markovian environment



Starting after a jump at time t_n at level x_n and state i_n, the level just after a new jump at time t_{n+1} is

$$x_{n+1} = p_{i_n}\left(x_n + \frac{t_{n+1} - t_n}{c_{i_n}}\right).$$

The next state i_{n+1} is selected randomly according to the distribution given by the i-th row of transition matrix $\boldsymbol{Q}$. Hence a first-step equation mixes:

- state-dependent linear growth $t \mapsto x + t/c_i$;
- state-dependent rescaling $x \mapsto x/p_i$;
- multiplication by selectors $\boldsymbol{E}_i$;
- matrix products with $\boldsymbol{Q}$.

Some notation

Let $\boldsymbol{z} = (z_i)$ and $\boldsymbol{\gamma} = (\gamma_i)$, with:

$$z_i = c_i(\lambda_i + w), \qquad \gamma_i = \frac{c_i \lambda_i}{p_i}.$$

Diagonal and selector matrices:

$$\boldsymbol{\mathcal{E}}(\boldsymbol{v}) = \mathbf{diag}(\mathrm{e}^{v_i}), \qquad \boldsymbol{\Delta}(\boldsymbol{v}) = \mathbf{diag}(v_i),$$

$$\boldsymbol{E}_i = (\delta_{ik}\delta_{i\ell})_{k\ell}.$$

Some useful matrices:

$$\boldsymbol{A}(s) = \boldsymbol{\Delta}\left(\frac{\boldsymbol{\gamma}}{\boldsymbol{z} - s}\right), \qquad \boldsymbol{B} = -\boldsymbol{\Delta}(\boldsymbol{\gamma})^{-1}.$$

Upward one-sided exit: the skip-free factorization

For $0 < x < a$, upward passage is continuous, so the process must cross x before a.

$$\boldsymbol{Z}(0 \uparrow a) = \boldsymbol{Z}(0 \uparrow x)\,\boldsymbol{Z}(x \uparrow a).$$

Consequently,

$$\boxed{\boldsymbol{Z}(x \uparrow a) = \boldsymbol{Z}(0 \uparrow x)^{-1}\boldsymbol{Z}(0 \uparrow a)}$$

and we are left with computing $\boldsymbol{Z}(0 \uparrow x)$ for any $x \geq 0$.

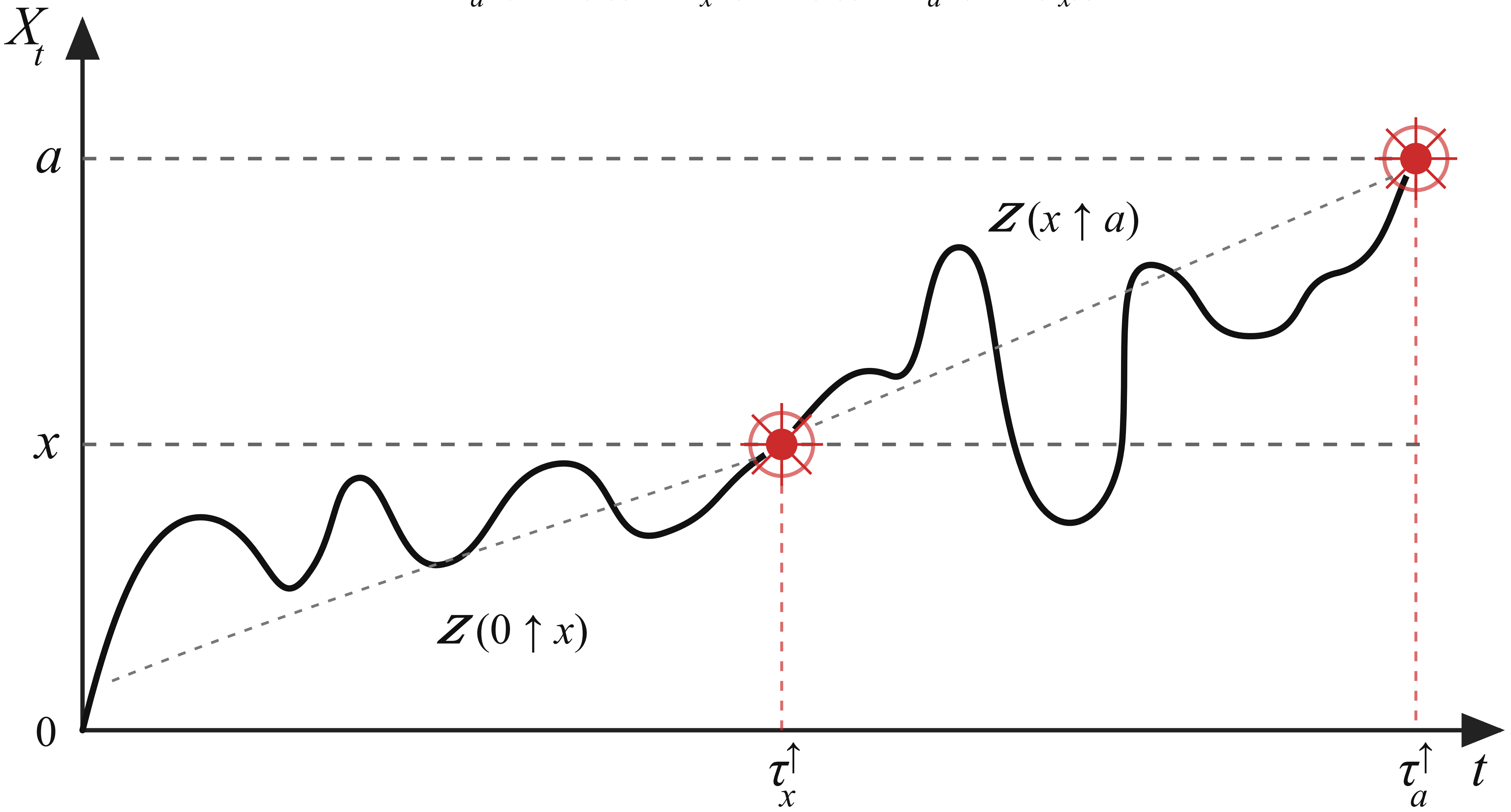
The proof is the strong Markov property at $\tau_x^{\uparrow}$, plus local invertibility of the matrix evolution.

Main ingredient

$$\tau_a^{\uparrow}(0, J(0)) \overset{\text{dist}}{=} \tau_x^{\uparrow}(0, J(0)) + \tau_a^{\uparrow}(x, J(\tau_x^{\uparrow}(0, J(0)))).$$

Strong Markov property at the hitting times

$$\tau_a^{\uparrow}(0, J(0)) \simeq \tau_x^{\uparrow}(0, J(0)) + \tau_a^{\uparrow}(x, J(\tau_x^{\uparrow}))$$



Computing the upward scale matrix

First-step analysis at the first jump gives

$$\boldsymbol{I} = \boldsymbol{\mathcal{E}}(-a\boldsymbol{z})\boldsymbol{Z}(0 \uparrow a)^{-1} + \boldsymbol{\Delta}(\boldsymbol{\gamma}) \int_0^\infty \boldsymbol{\Delta}(x \le a\boldsymbol{p})\boldsymbol{\mathcal{E}}(-(\boldsymbol{z}/\boldsymbol{p})x)\boldsymbol{Q}\,\boldsymbol{Z}(0 \uparrow a)^{-1}\,dx.$$

split at first jump $\Rightarrow$ Volterra-type matrix equation $\Rightarrow$ Laplace functional equation

Here $(x \le a\boldsymbol{p})_i = \mathbb{1}\{x \le ap_i\}$.

Upward one-sided exit: Laplace-domain solution

Let $\widetilde{\boldsymbol{U}}(s) = \int_0^\infty \mathrm{e}^{-sa}\boldsymbol{Z}(0 \uparrow a)^{-1}\, da$. Then

$$\widetilde{\boldsymbol{U}}(s) = \boldsymbol{A}(s)\boldsymbol{B} + \boldsymbol{A}(s)\sum_{i=1}^{m}\boldsymbol{E}_i\boldsymbol{Q}\,\widetilde{\boldsymbol{U}}(s/p_i).$$

Iterating this equation yields

$$\boxed{\widetilde{\boldsymbol{U}}(s) = \boldsymbol{A}(s)\boldsymbol{D}(s)\boldsymbol{B}}$$

$$\boldsymbol{D}(s) = \boldsymbol{I} + \sum_{n\geq 1}\sum_{i_1,\ldots,i_n}\boldsymbol{E}_{i_1}\boldsymbol{Q}\boldsymbol{A}(s/p_{i_1})\cdots\boldsymbol{E}_{i_n}\boldsymbol{Q}\boldsymbol{A}\left(\frac{s}{p_{i_1}\cdots p_{i_n}}\right).$$

The series converges because repeated multiplications by $p_i < 1$ push the residuals quickly towards 0.

Comparison with the scalar benchmark

Scalar AIMD

$$Z^{\uparrow}(w; x, a) = \frac{Z^{\uparrow}(w; 0, a)}{Z^{\uparrow}(w; 0, x)}$$

$$Z^{\uparrow}(w; 0, x) = \left[\sum_{k=0}^{\infty} \frac{(x/(w+\lambda))^k}{k!} (\lambda/(w+\lambda); p)_k\right]^{-1}.$$

Markov-modulated AIMD

Division is replaced by matrix inversion.

A scalar q-series is replaced by the non-commutative series $\boldsymbol{D}(s)$.

The matrices $\boldsymbol{E}_i\boldsymbol{Q}$ trace the regimes selected at each jump.

For $m = 1$, the matrix formula collapses to the scalar recursion from the previous paper.

From q-calculus: $(x; q)_n = (1-x)(1-xq)(1-xq^2)\cdots(1-xq^{n-1})$

Downward one-sided exit

For $x > b$, first-step analysis gives a renewal equation in the opposite direction:

$$Z_{ij}^{\downarrow}(w,x,b) = \sum_k q_{ik}\gamma_i \mathrm{e}^{z_i x} \int_{p_i x}^{\infty} Z_{kj}^{\downarrow}(w,y,b)\, \mathrm{e}^{-(z_i/p_i)y}\, dy, \qquad \boldsymbol{Z}^{\downarrow}(x,b) = \boldsymbol{I} \text{ for } x \le b.$$

With $\widetilde{\boldsymbol{Z}}^{\downarrow}(s) = \int_0^{\infty} \mathrm{e}^{-sx} \boldsymbol{Z}^{\downarrow}(x,b)\, dx$,

$$\widetilde{\boldsymbol{Z}}^{\downarrow}(s) = \sum_h \boldsymbol{A}(s)\boldsymbol{E}_h\boldsymbol{Q}\left(\widetilde{\boldsymbol{Z}}^{\downarrow}(s/p_h) - \widetilde{\boldsymbol{Z}}^{\downarrow}(z_h/p_h)\right).$$

$$\widetilde{\boldsymbol{Z}}^{\downarrow}(s) = \boldsymbol{A}(s)\boldsymbol{D}(s)\boldsymbol{H}, \qquad \boldsymbol{H} = -\sum_h \boldsymbol{E}_h\boldsymbol{Q}\widetilde{\boldsymbol{Z}}^{\downarrow}(z_h/p_h).$$

The matrix $\boldsymbol{H}$ is then recovered by solving a system of equations at the limiting arguments obtained for $s \to z_h/p_h$, $h \in E$.

Two-sided upward exit

Let $\boldsymbol{L}(x \uparrow_b^a)$ denote the transform of the upward exit time before downward exit.
We use a constructive recursion to compute it.

Assuming $\boldsymbol{L}(y \uparrow_b^a)$ known for all $y < a$, with $\rho < \min_i p_i$,
then $\boldsymbol{L}(x \uparrow_b^{a/\rho})$ is obtained for $x = a$ as

$$\boldsymbol{L}(a \uparrow_b^{a/\rho}) = (I - \boldsymbol{H}(a))^{-1} \boldsymbol{\mathcal{E}}(-(a/\rho - a)\boldsymbol{z}) \boldsymbol{L}(a \uparrow_b^a),$$

and for $x < a/\rho$ as

$$\boldsymbol{L}(x \uparrow_b^{a/\rho}) = \boldsymbol{\mathcal{E}}(-(a/\rho - x)\boldsymbol{z}) + \boldsymbol{H}(x) \boldsymbol{L}(a \uparrow_b^{a/\rho}).$$

The matrix $\boldsymbol{H}(x)$ with components $H_{ij}(x) = \gamma_i \mathrm{e}^{z_i x / p_i} \int_{p_i x}^{p_i a/\rho} \sum_h q_{ih} L_{hj}(w, y \uparrow_b^a) \mathrm{e}^{-(z_i/p_i) y}\, dy$ can be explicitly computed.

Two-sided upward exit — starting interval $[b, b/p)$

We have that, for any $x \in [b, b/p)$, where $p = \max_i p_i$, the following identity holds:

$$\boldsymbol{L}\left(x \uparrow_b^{b/p}\right) = \boldsymbol{\mathcal{E}}\big(-(b/p - x)\boldsymbol{p}\boldsymbol{\gamma}\big).$$

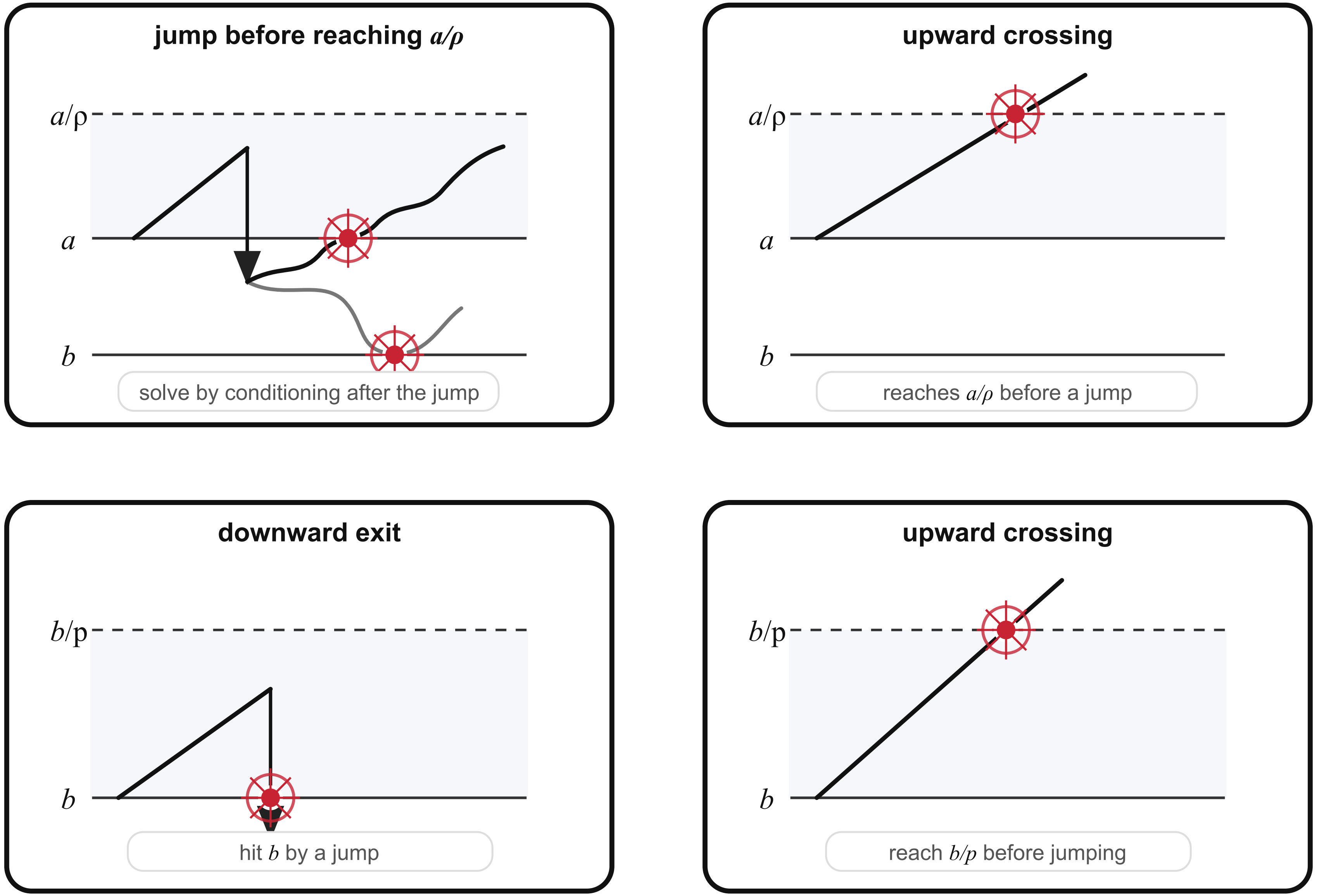
Proof.
Assume that $(X(0), J(0)) = (x, i)$, then since $yp_i < b$, for all $y \in [x, b/p)$, we have

$$\{\tau_{\uparrow b/p} < \tau_{\downarrow b}\} = \{T \geq c_i(b/p - x)\},$$

and

$$L_{ij}\left(x \uparrow_b^{b/p}\right) = \delta_{ij}\mathbb{P}_{x,i}(\tau_{\uparrow b/p} < \tau_{\downarrow b}) = \delta_{ij}\mathbb{P}_{x,i}(T \geq c_i(b/p - x)) = \delta_{ij}\exp\{-\lambda_i c_i(b/p - x)\}.$$

First-step decomposition for the AIMD exit problem



split the exit problem by conditioning on the first event in $[a,a/\rho]$ and $[b,b/p)$

Two-sided downward exit

Define

$$L_{ij}^{\downarrow}(w, x \downarrow_b^a) = \mathbb{E}_{x,i}\left[\mathrm{e}^{-w\tau_b^{\downarrow}}; \tau_b^{\downarrow} < \tau_a^{\uparrow},\, J(\tau_b^{\downarrow}) = j\right].$$

$$\boxed{L^{\downarrow}(w, x \downarrow_b^a) = Z^{\downarrow}(w, x, b) - L(w, x \uparrow_b^a) Z^{\downarrow}(w, a, b)}$$

Interpretation: to all paths that go to b, subtract paths that first go up to a and only then later hit b.

Reflected processes at its running infimum

2023 AIMD scalar model

They looked at the *upward exit time* for the process

$$X_t^- = \inf_{s \le t} X_s \wedge X_0,$$

that correspond to the free process reflected at it's minimum. The used technique is to show that the processes

$$e^{-w\tau_{a,b}(x)\wedge t} F\big(w, X_{\tau_{a,b}(x)\wedge t} \uparrow_b^a\big) - \int_0^{\tau_{a,b}(x)\wedge t} \mathcal{A}^\dagger F\big(w; X_s \uparrow_b^a\big)\, ds$$

is a local martingale, where

$$F(w, x \uparrow_b^a) = Z(w, x \downarrow_b) + L(w, x \uparrow_b^a)$$

Defining a similar object for the modulated version of the reflected process,

$$\boldsymbol{F}(x \uparrow_b^a) \approx \boldsymbol{Z}(x \downarrow_b) + \boldsymbol{L}(x \uparrow_b^a)$$

does not work.

The natural Markov modulated extension of $L(w, x \uparrow_b^a)$ is

$$\boldsymbol{L}(x \uparrow_b^{\boldsymbol{a}})$$

where $\boldsymbol{a}$ is a vector.

Working with state-dependent upper barriers

In general

$$\tau^{\uparrow}_{\boldsymbol{a}}(0, J(0)) \overset{\text{dist}}{\neq} \tau^{\uparrow}_{x}(0, J(0)) + \tau^{\uparrow}_{\boldsymbol{a}}(x, J(\tau^{\uparrow}_{x}(0, J(0)))),$$

that means

$$\boldsymbol{Z}(0 \uparrow \boldsymbol{a}) \neq \boldsymbol{Z}(0 \uparrow x)\, \boldsymbol{Z}(x \uparrow \boldsymbol{a}).$$

What instead works is

$$\tau^{\uparrow}_{\boldsymbol{a}}(0, J(0)) \overset{\text{dist}}{=} \tau^{\uparrow}_{x \wedge \boldsymbol{a}}(0, J(0)) + \tau^{\uparrow}_{\boldsymbol{a}}(x, J(\tau^{\uparrow}_{x \wedge \boldsymbol{a}}(0, J(0)))),$$

and

$$\boldsymbol{Z}(0 \uparrow \boldsymbol{a}) = \boldsymbol{Z}(0 \uparrow x \wedge \boldsymbol{a})\, \boldsymbol{Z}(x \uparrow \boldsymbol{a}).$$

Factorization over generated levels

For $x \leq \ell$,

$$\boldsymbol{Z}(x \uparrow \boldsymbol{a}) = \boldsymbol{Z}(x \uparrow \boldsymbol{a} \wedge \ell)\, \boldsymbol{Z}(\ell \uparrow \boldsymbol{a}).$$

Initialization

Let $\wedge \boldsymbol{a} = \min_i a_i$. For $x < \wedge \boldsymbol{a}$,

$$\boldsymbol{Z}(x \uparrow \boldsymbol{a} \wedge (\wedge \boldsymbol{a})) = \boldsymbol{Z}(x \uparrow \wedge \boldsymbol{a}),$$

which is the fixed-barrier problem already solved.

Iteration

At each new level, solve only for the active regimes.

Inactive regimes contribute with an identity matrix.

Extension step for modulated barriers

Assume $\boldsymbol{Z}(y \uparrow \boldsymbol{a} \wedge \ell)$ is known for $y < \ell$. Choose ρ so that $p_i x < \ell$ for $x \in [\ell, \ell/\rho)$.

$$\boldsymbol{Z}(\ell \uparrow \boldsymbol{a} \wedge (\ell/\rho)) = (\boldsymbol{I} - \boldsymbol{E}_k \boldsymbol{H}(\ell))^{-1} [\boldsymbol{I} + \boldsymbol{E}_k (\boldsymbol{\mathcal{E}}(-(\ell/\rho - \ell)\boldsymbol{z}) - \boldsymbol{I})],$$

$$\boldsymbol{Z}(x \uparrow \boldsymbol{a} \wedge (\ell/\rho)) = \boldsymbol{E}_k \boldsymbol{\mathcal{E}}(-(\ell/\rho - x)\boldsymbol{z}) + \boldsymbol{H}(x)\boldsymbol{Z}(\ell \uparrow \boldsymbol{a} \wedge (\ell/\rho)) + (\boldsymbol{I} - \boldsymbol{E}_k).$$

$$H_{ij}(x) = \gamma_i \mathrm{e}^{z_i x/p_i} \int_{p_i x}^{p_i \ell/\rho} \sum_h q_{ih} Z_{hj}(w, y \uparrow \boldsymbol{a} \wedge \ell) \mathrm{e}^{-(z_i/p_i)y}\, dy.$$

Take-home points

1

Upward skip-free motion still gives clean multiplicative identities.

2

First-step analysis yields matrix-functional equations with rescaling $s \mapsto s/p_i$.

3

The scalar scale-function paradigm becomes a matrix scale-function paradigm.

Questions

Thank you.

Bernardo D'Auria, University of Padova

Joint work with Zbigniew Palmowski

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Boundary conventions

Upward crossing is by creeping:

$$X(\tau_a^{\uparrow}) = a.$$

Downward crossing is by a jump:

$$X(\tau_b^{\downarrow}) \leq b.$$

This asymmetry is why upward identities factor cleanly, while downward identities require integral renewal equations.